Agentic AI for Economic Research

Practical workshop for first-year PhD students in Economics

Format: 4 sessions × 2 hours (8 hours total)
Instructor: Prof. Marco Le Moglie – Catholic University of the Sacred Heart

Why this course

Economic research still runs on the same steps—literature, data, identification, estimation, writing, revision—but the tools around those steps are changing quickly. Agents can now read files, run code, search the web, and carry out multi-step tasks on your behalf. That can save real time, but only if you know how to set them up, what to delegate, and what to verify.

This course is not about whether AI will replace economists, nor about choosing the "best" model. It is a practical introduction to *agentic workflows*: how to use AI assistants as controlled collaborators across a research pipeline, without losing oversight of what they do.

Structure

Before class. The slides and the demo folders in the course repository are the materials. Come to Session 1 with one harness installed: ChatGPT desktop with Codex, Claude Code, Cursor, or OpenCode.

Session	Topic
1	**Vocabulary.** What the pieces are, and what they are not. A model writes the next bit of text. The harness holds the conversation, calls tools, and stops for permission. Context, tokens, and what happens when the window fills. Rules, MCP, and skills. Which model you call, what a long run costs, and which data stays off a consumer chat. In class: a Stata do-file with a bug left in on purpose, a skill the agent should load, and a short pull of public municipal data.
2	**Second brain.** How the agent keeps this project the next time you open it. We start from JARVIS, the film assistant, and from a clip of that assistant being built with an agent. Then the choices: paste the files, search them, keep a wiki the agent updates, or a hybrid of project files plus search. We build the small version in Obsidian, and I show the Brain I use, including the Sunday pass that checks it. The demo continues from Session 1 into that vault, and into the same folder on Dropbox, Overleaf, and the phone.
3	**Automations.** How the same checks run when you are not in the session. A schedule ingests the notes, updates the project files, and leaves a log. The Sunday pass from Session 2 is the picture: the agent reads the vault, moves finished work into the log, and asks you the questions it cannot close. Slides not yet written.
4	**Control.** How you check the work before you trust it. The diff, the table, and the log. Session 1 already ends on a table that runs and still needs a reader. This session makes that check a habit: what the agent may change, what you must read, and what never becomes permanent without you. Slides not yet written.

What you will be able to do

By the end of the course, you should be able to:

1. **Name** the pieces: model, harness, context, tool, permission, skill.
2. **Point** an agent at a project folder and say what it may change.
3. **Keep** the project in files the next session can open, instead of in the chat.
4. **Leave** a repeated check running when you are not in the session.
5. **Read** the diff, the table, and the log, and decide what is allowed to stay.

What this course is not Not a programming course (basic familiarity with Stata/R/Python and LaTeX is enough); not a survey of every AI product on the market; not a substitute for econometric training or research judgment.

Pre-course readings, slide sources, and session recordings: github.com/MarcoLeMoglie/Agentic_Economic_Research.